Familiarization with Scientific and Information Visualization methods

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I. Introduction

This project consists of various tasks related to scientific and information visualizations.

A. Scientific Visualization

The three visualization tasks in the scientific visualization are based on gridMET dataset [1].

1) color mapping
2) contour mapping
3) quiver plots

B. Information Visualization

The tasks in information visualization are based on the US Accidents dataset [2].

1) Node-link diagrams
2) Parallel coordinates plot with edge bundling and zoomable treemap visualizations.
3) horizon plots and ridgeline plots

C. Work Distribution

Scientific Visualization

- color mapping - Dayanithaa.
- contour mapping - Sahasara.
- quiver plots - Ashwini.

Information Visualization

- Parallel coordinates plot with edge bundling and zoomable treemap visualizations. - Dayanithaa.
- horizon plots and ridgeline plots - Sahasara.
- Node-link diagrams - Ashwini.

II. Scientific Visualization

gridMET [1] is a dataset of daily high-spatial resolution (4-km, 1/24th degree) surface meteorological data covering the contiguous US from 1979-yesterday. These data can provide important inputs for ecological, agricultural, and hydrological models. These data are updated daily. gridMET grids correspond to the centroid of the 4-km x 4-km pixel. The Primary Climate Variables captured are Maximum temperature, minimum temperature, precipitation accumulation, downward surface shortwave radiation, wind-velocity, humidity (maximum and minimum relative humidity and specific humidity. This dataset is used to perform the three scientific visualization tasks.

A. color mapping

1) Dataset: In this section, the variables used for visualization are:

- Daily Maximum Air Temperature (`tmmx`): Represents maximum daily temperature and helps analyze spatial and temporal temperature variations.
- Daily Precipitation (`pr`): Represents daily precipitation and helps examine the spatial distribution and intensity of rainfall.
- Daily Solar Radiation (`srad`): Represents incoming solar radiation and helps analyze variations in solar energy received across different geographical regions.

These variables were selected to study different atmospheric and environmental characteristics over the region. Temperature captures thermal variations, precipitation represents rainfall distribution, and solar radiation provides insight into the spatial variation of incoming solar energy.

2) Data Preprocessing: The following preprocessing steps were performed to prepare the gridMET dataset for colour mapping experiments:

- Loading the dataset: The NetCDF files were loaded using the `xarray` library. The dataset was examined to understand its dimensions, coordinates, data variables, and overall structure.
- Variable examination: The relevant meteorological variables, namely daily maximum temperature (`tmmx`), precipitation (`pr`), and solar radiation (`srad`), were identified and extracted for further analysis.
- Attribute inspection: The metadata and attributes associated with each variable were examined to understand their descriptions, measurement units, and other relevant information.
- Missing value analysis: The selected variables were checked for missing or invalid values to assess data completeness and identify any preprocessing requirements before visualization.
- Summary statistics: Descriptive statistics, including minimum, maximum, mean, and standard deviation, were computed for each variable to understand their value ranges, distributions, and variability.

- Temporal extraction: A contiguous three-month period from 1 July to 30 September 2021 (92 days) was extracted from the complete dataset for analysis.
- Temporal sampling: Five representative dates were sampled uniformly across the selected three-month period, including the first and last dates (1 July and 30 September 2021). These samples were used to visualize and compare changes in the spatial distribution of the selected variables over time.
- Range computation: Global minimum and maximum values were determined across the selected time instances for global normalization, while individual minimum and maximum values were computed for each time instance for local normalization.
- Handling precipitation values: Zero precipitation values were masked before applying logarithmic normalization, since the logarithm of zero is undefined.

3) Methodology: Colour mapping involves two important steps: normalization and colour assignment. Normalization transforms the original scalar values into a suitable range, after which a colour map assigns a corresponding colour to each value.

- Global Normalization: Global normalization uses a common minimum and maximum value computed across all selected time instances. This ensures that the same scalar value is represented by the same colour across different dates.
  It is particularly useful for comparing absolute values over time and identifying temporal changes in the spatial distribution of a variable.
- Local Normalization: Local normalization calculates the minimum and maximum values independently for each time instance. This emphasizes relative spatial variations within individual dates.
  However, the same colour may represent different absolute values across different dates, making direct temporal comparison less reliable.
- Discrete Colour Mapping: Discrete colour mapping divides a continuous range of numerical values into a fixed number of intervals, with each interval assigned a distinct colour. This makes it easier to distinguish different value ranges and identify regions belonging to particular categories.
- Logarithmic Normalization: Logarithmic normalization applies a logarithmic transformation to scalar values before assigning colours. It is particularly useful for datasets with highly skewed distributions, where a small number of large values dominate the data.
  Precipitation values often exhibit this characteristic. Logarithmic normalization improves the visibility of variations among smaller positive values. Since the logarithm of zero is undefined, zero precipitation values must be masked before applying this technique.

Observations

- Clear seasonal shift from July to September. Solar radiation falls steadily, with 30 Sep the lowest of the five dates. Maximum temperature also cools by 30 Sep, mainly in the north and northeast. September precipitation is below average over most of the country.
- The Southwest desert is consistently hot, clear and dry. It has the highest maximum temperatures, the lowest solar variability, low precipitation, and it is the persistent hotspot on all five dates.
- Precipitation is very uneven. Most of the area is low and a few cells are very high, which is why the log scale shows more detail than the equal-bin version.

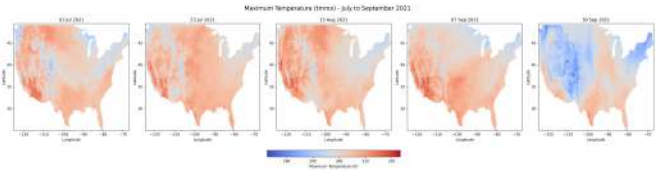


Fig. 1. Maximum temperature colour maps for five representative dates from July–September 2021.(global)

Inference: The map shows that the Southwestern regions s and the Central Valley are the hottest regions, while mountain areas are cooler because of altitude. By 30 September the north and Northeast have cooled a lot, while the south stays warm. Overall, we could come across higher temperatures during Mid August and lower temperatures by the end of September. This is a global colour map.

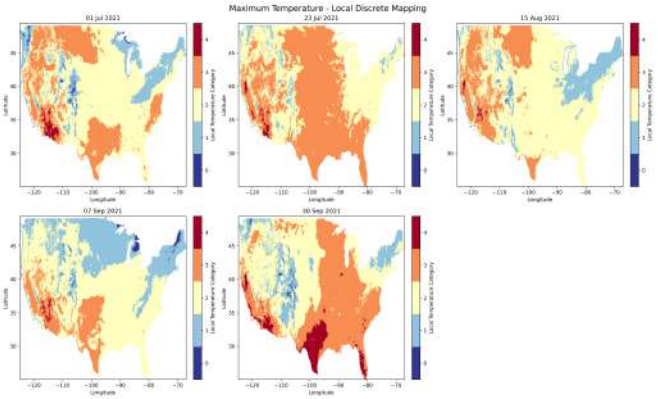


Fig. 2. Maximum temperature colour maps(local discrete mapping)

Inference: The map is a colour map that uses local normalisation and discrete colour mapping to visualise generally hotter regions and colder regions across the country. We can observe a few areas that consistently maintain a high temperature and few that consistently maintain a lower temperature.The northern United States, and northeastern states, generally maintain lower temperature categories across the observed dates. These patterns suggest a relatively persistent geographical distinction, with warmer conditions concentrated in the southern and southwestern regions and cooler conditions prevailing across the northern parts of the country.

Fig 3 and Fig 4 represent persistent hotspots that constantly maintain high temperatures. Fig3 compares various sequen-

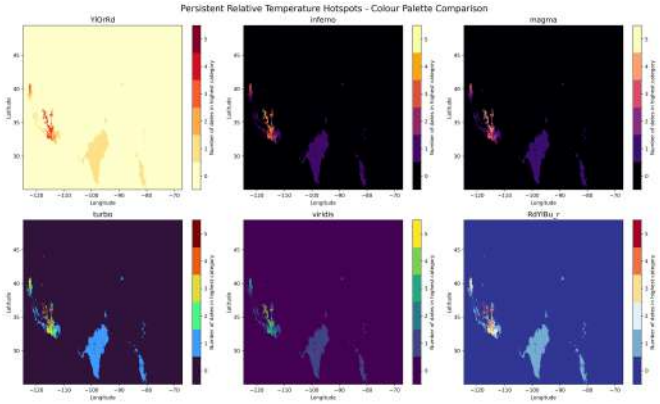


Fig. 3. Palette comparison for hotspots

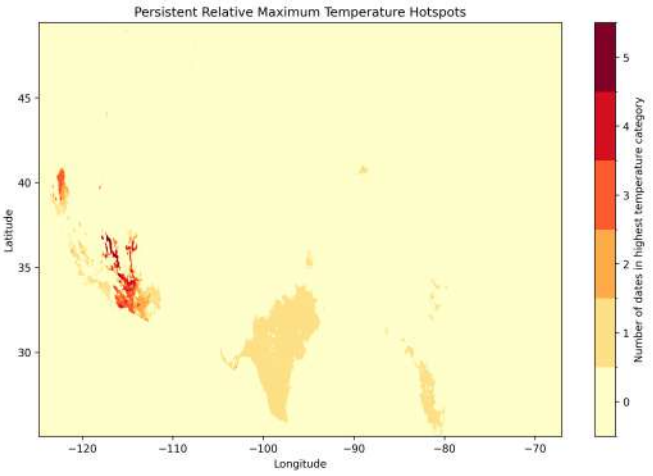


Fig. 4. Persistent hotspots

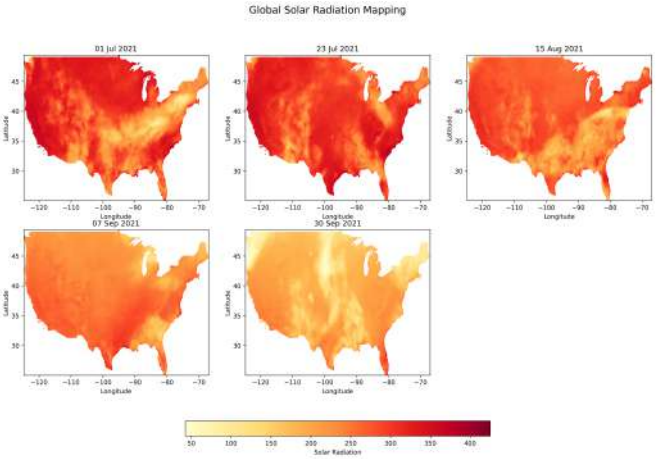


Fig. 5. Contour maps of solar radiation(srad) (global mapping)

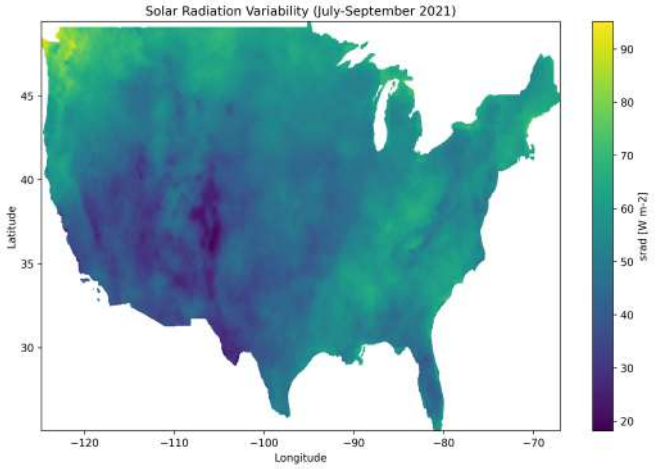


Fig. 6. Variability of solar radiation across regions

tial colour maps to see which can be perceived easily. The persistent hotspot analysis reveals that the highest relative maximum temperatures are consistently concentrated in the western United States.These regions exhibit the highest hotspot counts, indicating that they repeatedly fall within the highest temperature category across the sampled dates.

Fig 5 shows the global solar radiation map across 5 dates. The global solar radiation maps show relatively high radiation levels across much of the United States during July and August, with particularly elevated values across the western and northern regions. The southern regions, including Texas and parts of the Gulf Coast, also exhibit high radiation levels. By September 7, radiation levels begin to decline across several regions, and this reduction becomes more evident by September 30, when much of the country displays comparatively lower values. Overall, the maps illustrate a gradual seasonal decline in solar radiation from summer towards autumn.

Fig 6 shows the variability map of solar radiation that indicates the amount the variable varies across regions of the country. The variability map highlights substantial spatial differences in solar radiation across the United States during the study period. Higher variability is observed across the northwestern regions and parts of the eastern United States, as indicated by higher values on the colour scale. In contrast, relatively lower variability is concentrated across the southwestern and south-central regions, including parts of Arizona, New Mexico and Texas. This suggests that solar radiation remains comparatively stable across these regions, whereas the northwestern and eastern regions experience greater fluctuations over the study period.

Fig 7 depicts the difference between the actual values when compared to the average value of the attribute in that geographic region. The anomaly maps reveal considerable spatial and temporal variations in solar radiation across the United States. In July and August, positive anomalies (orange/red) are observed across much of the northern and western regions, while negative anomalies (blue) are concentrated along parts of the central and southeastern regions. By September 7, negative anomalies become more widespread, particularly across the northern and central United States. This pattern becomes even more pronounced by September 30, when much of the

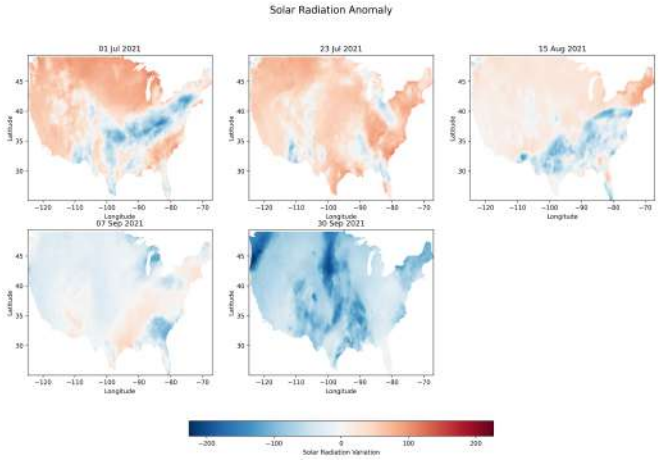


Fig. 7. Variation from average value

country exhibits negative anomalies, indicating a substantial reduction in solar radiation relative to the reference values.

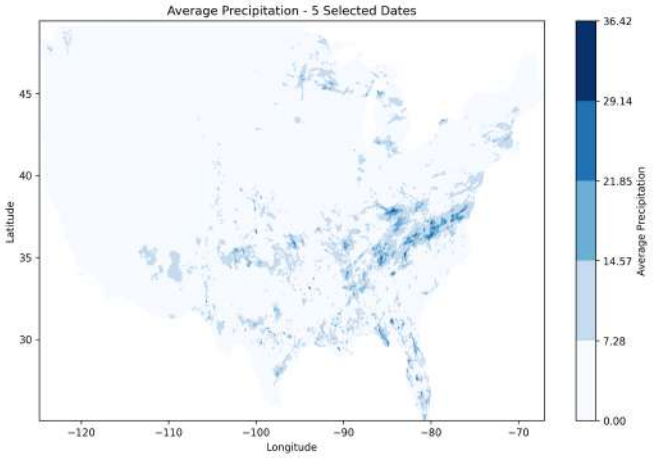


Fig. 8. Average precipitation across regions

Fig 8 shows the average precipitation. The average precipitation map exhibits a highly skewed distribution, with most grid cells recording very low or zero precipitation values, while only a relatively small number of regions experience substantially higher precipitation. This large difference in magnitude makes linear colour mapping ineffective, as lower values become difficult to distinguish. Therefore, logarithmic colour mapping is used to enhance the visibility of variations among smaller precipitation values while still representing regions with higher precipitation levels. This allows spatial patterns and regional differences to be identified more clearly despite the highly uneven distribution.

Fig 9 shows the logarithmic mapping of precipitation. The logarithmic colour mapping highlights the wide spatial variation in average precipitation across the United States. Higher precipitation levels are observed across the eastern and southeastern regions, particularly around the Great Lakes and parts of the eastern coastline. The western and southwestern regions

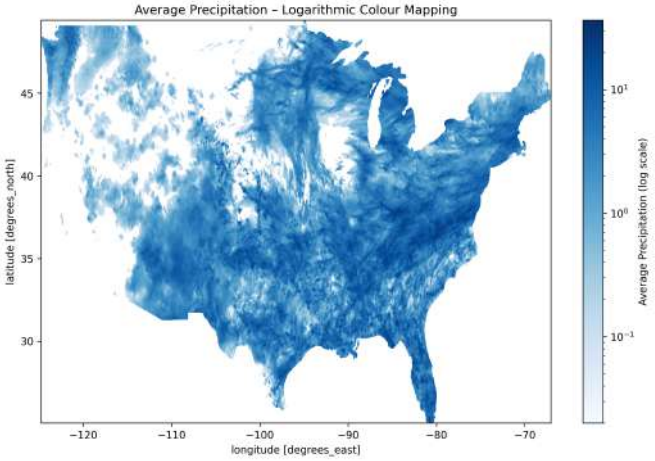


Fig. 9. Logarithmic colour mapping of precipitation

generally exhibit lower precipitation levels, with relatively dry conditions across parts of the Great Basin and surrounding areas. The logarithmic scale helps reveal variations across regions with significantly different precipitation magnitudes, making smaller precipitation values more distinguishable.

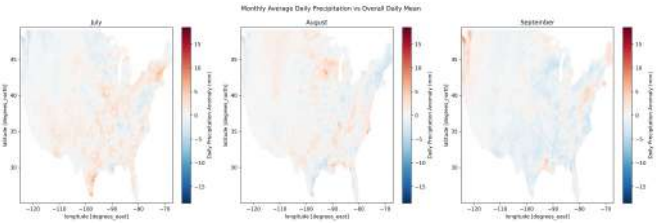


Fig. 10. Variation from average per month

The monthly anomaly maps reveal changes in precipitation relative to the overall daily mean. July and August exhibit a mixture of positive and negative anomalies, with positive anomalies appearing across parts of the northern and central regions and negative anomalies scattered across the western and eastern United States. By September, negative anomalies become more widespread across much of the central and southern regions, suggesting comparatively drier conditions than the overall average. However, certain areas in the northwestern and northeastern regions continue to exhibit positive anomalies, indicating localized increases in precipitation.

4) Discussion: Overall, the data show a clear seasonal change from July to September, with solar radiation and temperature falling and September being drier across most of the country. The Southwest is consistently hot, clear and dry, while the Pacific Northwest changes the most through the season. Cloud cover links the variables, with low-radiation areas matching wetter regions. Terrain also shapes the patterns, as mountains are cooler and the interior West is dry. Precipitation is uneven, and persistent temperature hotspots are rare and limited to the Southwestern regions.

B. Contour mapping

1) Dataset: In this section, the variables used for visualization are:

- Daily Maximum Air Temperature (`tmmx`)
- Daily Minimum Air Temperature (`tmmn`)
- Mean Vapor Pressure (`vp`)
- Maximum Relative Humidity (`rmax`)

These variables were selected to analyze different aspects of atmospheric conditions over the region. The temperature variables (`tmmx` and `tmmn`) capture daytime and nighttime thermal variations, while mean vapor pressure (`vp`) represents the amount of moisture present in the air, and maximum relative humidity (`rmax`) indicates the highest daily humidity levels. Contour plots are used to visualize the spatial distribution of these continuous variables, enabling the identification of regional patterns, gradients, and seasonal variations.

The slice analysed is 1 July–30 September 2021 (92 days). three-month summer slice was used because it contains both the seasonal peak and the start of autumn cooling.

2) Data Preprocessing: The preprocessing pipeline consisted of the following steps:

- The required NetCDF files were loaded using the `xarray` library.
- A contiguous three-month period (July–September 2021) was extracted from the complete yearly dataset for analysis.
- Temperature values (`tmmx` and `tmmn`) were converted from Kelvin to degrees Celsius using

$$T_{\circ C} = T_K - 273.15.$$

- The latitude and longitude coordinate arrays were extracted to preserve the spatial structure required for contour visualization.
- Representative temporal samples were selected from the three-month period for visualization. Both uniformly sampled dates and event-based dates were considered to capture seasonal variation and significant weather patterns.
- Global and local contour levels were computed depending on the visualization. Global contour levels maintain a consistent scale across all time steps for comparison, while local contour levels adapt to the range of each individual frame to highlight finer spatial variations.

3) Methodology: Contour lines (isolines) were drawn with Matplotlib `contour` and filled bands with `contourf`, which build on a marching-squares-type algorithm [**?**], [**?**]: each grid cell is classified by which corners lie above the isovalue, and the crossing is placed by linear interpolation along the cell edges. Both were used because they answer different questions: isolines give exact values and show gradient (line spacing), while filled contours give fast area-wise reading. Final maps combine a fine fill (161 levels, 0.25 °C steps) with sparse 5 °C labelled lines, drawn on a PlateCarree map with state and coast outlines (Cartopy).

Time instances Nine dates were sampled uniformly (every 11–12 days, both endpoints included): 1, 12, 23 Jul; 4, 15, 26 Aug; 7, 18, 30 Sep. Uniform sampling covers the season without bias

4) Experiments: the following experiments were performed to decide the factors

Number of levels experimented with With 5, 10, 20 and 40 .A moderate number of contour levels (30) was selected to balance visual clarity and detail. Using too few levels oversimplifies the spatial variation and may hide important temperature gradients, whereas too many levels produce densely packed contour lines that make the visualization difficult to interpret.

Equal-interval versus quantile levels Equal-interval contour levels were used because they provide a consistent representation of changes in the data and make the contour spacing easier to interpret. Quantile-based levels were also experimented with, but they tend to concentrate contour levels in regions with dense data values, making comparisons less intuitive. Therefore, equal-interval levels were selected for the final visualizations.

Smoothing A NaN-aware Gaussian filter was applied to reduce noise and remove small, isolated contour loops while preserving the overall spatial patterns. The smoothed data were used to generate all final contour visualizations.

Discrete vs. continuous color fill. Continuous color fill was selected for the final visualizations as it provides smoother transitions and better represents gradual spatial variations. Contour lines were overlaid to improve readability and allow easy interpretation of the contour values.

5) Plots: All maps use the smoothed field (Gaussian filter, $\sigma = 1$ cell), a continuous fill of 161 levels with sparse labelled contour lines, and state and coast outlines. Each plot is described below before its observations.

Fig. 11 shows the daily maximum temperature (`tmmx`, °C) for the nine uniformly sampled dates. All nine maps use one common colour scale of 10–50 °C (the `YlOrRd` colormap, values outside the range shown with `extend="both"`), with labelled lines every 5 °C. Because the scale is shared, the colours can be compared directly between dates.

Observations (Fig. 11)

- Persistent high-temperature contours are observed over the southwestern United States, while cooler contour bands remain confined to the Rocky Mountain region.
- During August, high-temperature contours expand eastward, covering a larger fraction of the central United States compared to July.
- By late September (30 September), the contour field contracts, with lower-temperature contours extending southward, indicating the onset of seasonal cooling.
- Strong contour gradients are observed near mountainous regions, reflecting the influence of topographic elevation on surface temperature distribution.
- The global contour levels maintain a consistent mapping between colour and temperature, enabling reliable temporal comparison across all selected dates.

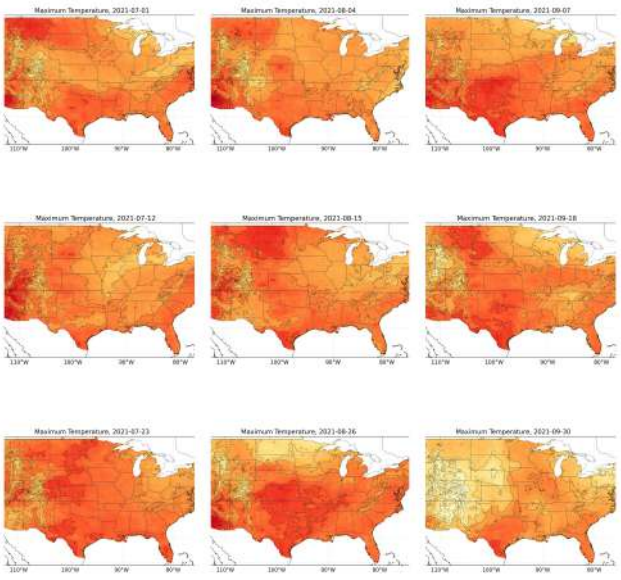


Fig. 11. Daily maximum temperature (`tmmx`, °C) for nine uniformly sampled dates, 1 July–30 September 2021, on one common global colour scale (10–50 °C); lines every 5 °C.

Fig. 12 repeats the same nine dates and the same fill and line settings, but with local scaling: the colour range and the line levels of each map come from that day's own minimum and maximum.

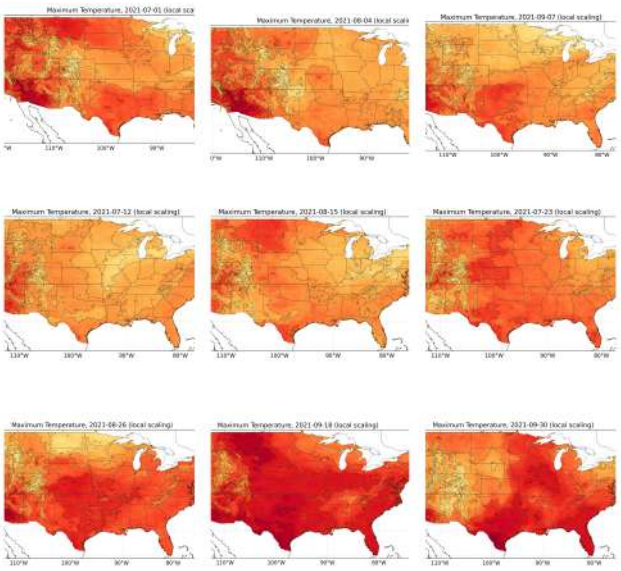


Fig. 12. Daily maximum temperature (`tmmx`, °C) for the same nine dates as Fig. 11, each map scaled to its own minimum and maximum (local scaling); colours are not comparable between panels.

Summary (Fig. 12) Local scaling emphasizes intra-day spatial temperature variations by enhancing regional thermal gradients and localized contour features. While global scaling is better suited for inter-date comparison, local scaling provides a clearer view of fine-scale spatial structures and regional anomalies, making both approaches complementary for temperature analysis.

Fig. 13 shows the diurnal temperature range (DTR), computed for every grid cell as the daily maximum minus the daily minimum temperature (`tmmx` − `tmmn`, both in °C). It is mapped for the first day of each month (1 July, 1 August and 1 September 2021) to show the month-to-month change. The colour range is taken from the 0.5th–99.5th percentile of DTR over the season, so that a few extreme cells do not stretch the scale, with lines every 5 °C.

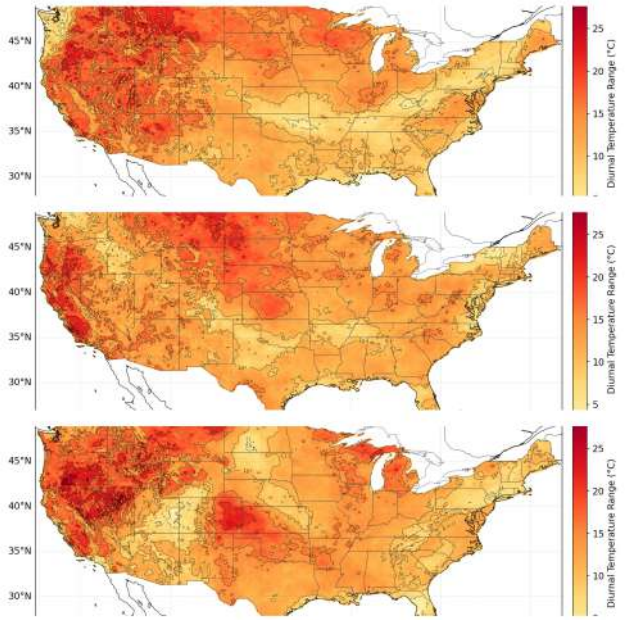


Fig. 13. Diurnal temperature range (`tmmx` − `tmmn`, °C) over the continental United States on 1 July, 1 August and 1 September 2021; lines every 5 °C.

Observations (Fig. 13)

- Higher DTR values are observed over the western United States during all three months, while the eastern and southeastern regions show comparatively lower DTR.
- The western mountainous regions show closely spaced contours, indicating rapid spatial changes in DTR due to changes in elevation.
- In August, moderate DTR values spread across a larger part of the northern and central United States compared to July.
- By September, higher DTR values become more prominent over the central High Plains, while the eastern coastal regions continue to show relatively low DTR.
- The contour patterns show clear spatial and seasonal variations in diurnal temperature range from July to September.

Fig. 14 shows the maximum relative humidity (`rmax`, %) for the same three dates. A blue sequential colormap (`YlGnBu`) was used because humidity is a different quantity

from temperature. The colour range again comes from the 0.5th–99.5th percentile of the season, with lines every 10 %.

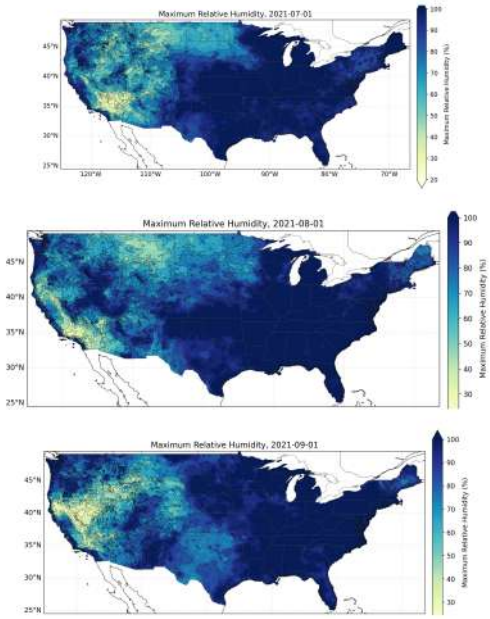


Fig. 14. Maximum relative humidity (`rmax`, %) over the continental United States on 1 July, 1 August and 1 September 2021; lines every 10 %.

Observations (Fig. 14)

- The eastern United States maintains high maximum relative humidity (80–100%) throughout all three months, while the western region shows lower humidity.
- August shows a slight increase in humidity over parts of the western interior compared to July.
- Areas with higher maximum temperatures generally have lower relative humidity, whereas cooler regions exhibit higher humidity.

Fig. 15 shows the vapor pressure deficit (VPD, kPa), which measures how much more moisture the air could hold. It is mapped for 1 July, 1 August and 1 September 2021 with the same method as above: colour range from the 0.5th–99.5th percentile of the season, the `YlOrRd` colormap and lines every 1 kPa.

Observations (Fig. 15)

- Highest VPD occurs over the southwestern United States, while the eastern region has low VPD.
- High-VPD regions expand in August and decrease slightly by September.
- Higher maximum temperatures generally correspond to higher VPD.

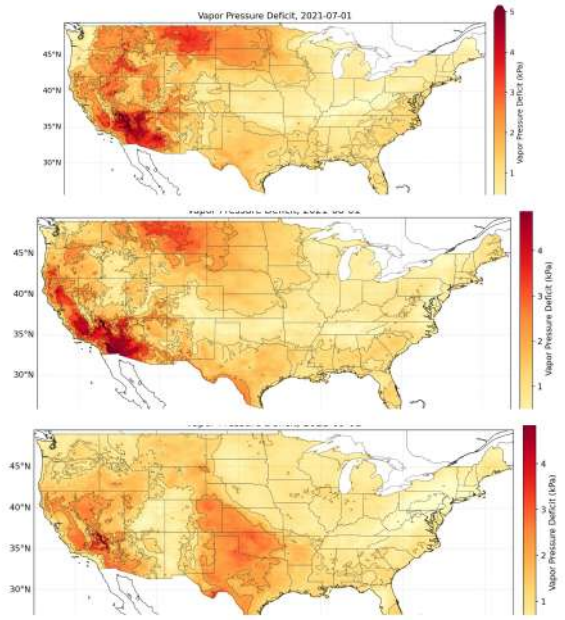


Fig. 15. Vapor pressure deficit (VPD, kPa) over the continental United States on 1 July, 1 August and 1 September 2021; lines every 1 kPa.

6) Discussion:: The contour maps reveal strong spatial relationships among the selected meteorological variables. Regions with higher maximum temperature generally coincide with higher vapor pressure deficit (VPD), indicating increased atmospheric moisture demand. In contrast, maximum relative humidity exhibits an inverse spatial pattern, with humid regions corresponding to lower VPD. Areas with high diurnal temperature range (DTR) are primarily associated with the dry western interior, where low humidity promotes larger day–night temperature variations. Together, these contour visualizations effectively capture the coupled behaviour of temperature, humidity, and atmospheric moisture across the study period.

C. Quiver Plot

Quiver plot is a visualization method to view a vector field. It represents both direction and magnitude of the vector using arrows at various points in 2D or 3D space. Here quiver plot is used to visualize wind velocity over United States(US) for a time duration of 3 months.

1) Dataset: The wind velocity for 3 months starting from July 2021 to September 2021 over US was obtained from gridMET dataset. The data in gridMET dataset is available as .netCDF files. The two .netCDF files th_2021.nc and vs_2021.nc containing wind related meteorological data for the year 2021 are downloaded. th_2021.nc dataset contains information about daily mean wind speed in meter/second. The data have both temporal and spatial attributes, the latitude and longitude of the place at which the wind speed is recorded and the day of the year. Similarly vs_2021.nc dataset contains records on daily mean wind direction clockwise from north along with details of latitude, longitude of the place. Records

specific to period July 2021 to September 2021 are extracted from the th_2021.nc and vs_2021.nc files.

2) Preprocessing: As wind speed and wind direction are taken from two different files, the data slices from both files are checked for the same grid structure of latitude, longitude and time and then the slices are merged to form one single uniform dataset containing wind speed and wind direction over US for July 2021 to September 2021 time frame. Minimum and maximum values of wind speed and wind direction are inspected for validity of the data. The max value for wind direction should be less than or equal to 360 degree as the wind direction is circular. The wind speed values are kept between valid 0 m/s to 60 m/s by removing the extreme values and bad data if any. Wind direction being circular, wind direction values are wrapped between 0 to 360 degrees and kept only for valid wind speed. For calm days, wind speed of 0, wind direction was removed. In order to convert the wind speed and wind direction in wind velocity vector, the polar coordinates are converted to U and V vectors signifying eastward and northward wind vectors respectively. Once we get the wind velocity in terms of U and V wind vectors, it is easy to plot wind velocity in 2D space.

3) Implementation: The python library "xarray" specifically designed to work with multi-dimentaional arrays is used along with numpy library to extract and preprocess data from the netCDF files. Matplotlib's plotting interface pyplot is used to plot all the quiver plots of wind velocity. Cartopy is a python library specially designed for geographical maps. Cartopy has capabilities to add latitude, longitude projections, map features like coastlines,country borders, rivers and geographic gridlines and labels. Cartopy library is used along with pyplot to plot wind velocity quiver plots of the country of United states.

4) Quiver Plot: Fig. 16 Shows map of mean wind velocity of 14 days from 1st July to 14th July 2021, over United States. Using cartopy and pyplot library the quiver plot is plotted. The map has latitude, longitude labels (°W, °N), dashed grid lines, state borders and a colorbar from 0 to about 4.4 m/s indicating the various wind velocity. The arrow glyph with 25 step size is used to show the wind velocity at every 25th cell in the whole grid,so approximately 1° between 2 arrows with arrow length proportional to the wind speed. The "2 m/s" key at the bottom right indicates the arrow length w.r.t to 2 m/s speed. The speed of wind at various locations is indicated using colormap, the color legend at right indicates various ranges of wind speed associated with colors. As can be seen with this sample, with 25 step size the long arrows are overlapping with each other and resulting in a cluttery visualization. The visual clutter can be reduced to make map more readable by using grid sampling. Grid sampling is a method where a small subset of cells from large grid are used to make the resulting visualization more aesthetic and easy to analyze. Another approach to make cleaner visualization specific to quiver plot is length of the arrow glyph. The arrow length can be magnitude proportional or uniform. In case of uniform arrow length, the magnitude of arrow can be represented using some other channel e.g. color. Below are the two experiments

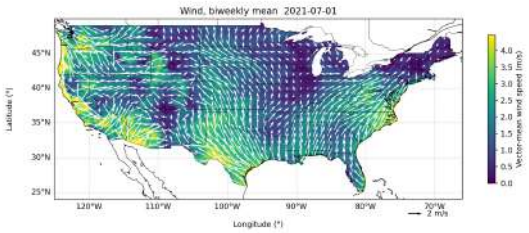


Fig. 16. Wind Velocity Over United States biweekly Mean starting 2021-07-01

carried out using two grid sampling methods along with arrow sizes over the wind velocity data.

5) Stride Grid Sampling: Stride sampling is a point sampling method where every arrow represents value of the cell. It is faster rendering method and keeps real cell values. The drawback of this approach is it does not represent its neighborhood so simply by changing the starting point and step, the whole visualization can change. In case of NaN value at a cell, the arrow is dropped.

Fig. 17 shows quiver plots drawn for three different step sizes for biweekly mean wind velocity over united States starting from 1st July 2021. The layout, color scale, and key are identical, but each arrow comes from a single grid cell. For the three rows, step size is varied between every 10th, 20th and 40th cell from top row to bottom row respectively. The left column has arrows drawn proportional to the magnitude of wind speed value at that cell along with colors representing wind speed redundantly while the right column has arrows drawn with uniform length and the magnitude of the wind is shown using colormap. The arrows are scaled with respect to space between cells or the step size. The total arrows in the grid for 10th, 20th and 40th step cell are 4863, 1202, and 297 respectively.

In case of step size 10, spacing of 10 cells between 2 arrows, the map has highest no. of arrows. For the magnitude proportional arrow length, the arrows length is scaled according to step size, thus the arrow with hightest wind speed occupying full space and weaker winds producing shorter arrows. The uniform length arrows are drawn as unit vectors with length 80% of the spacing. As seen from figure, the smaller step size of 10 with magnitude proportional arrows produces very tiny arrows for weaker winds resulting in a sparse visualization while with uniform length arrows the visualization becomes very dense. In both cases it is very difficult to analyze the map and interpret a pattern. In case of step size 20, both the magnitude proportional and uniform arrow length maps are less sparse and less dense respectively compared to step size 10 and wind pattern is clearly visible. In case of step size 40, the magnitude proportional arrows map is sparser than step size 10 magnitude proportional arrows map but still tiny arrows for weaker winds make it difficult to interpret any pattern. For step size 40 uniform arrows map, the map is sparser than step size 20 uniform arrows map which hints of some wind pattern but unable to represent clearly.

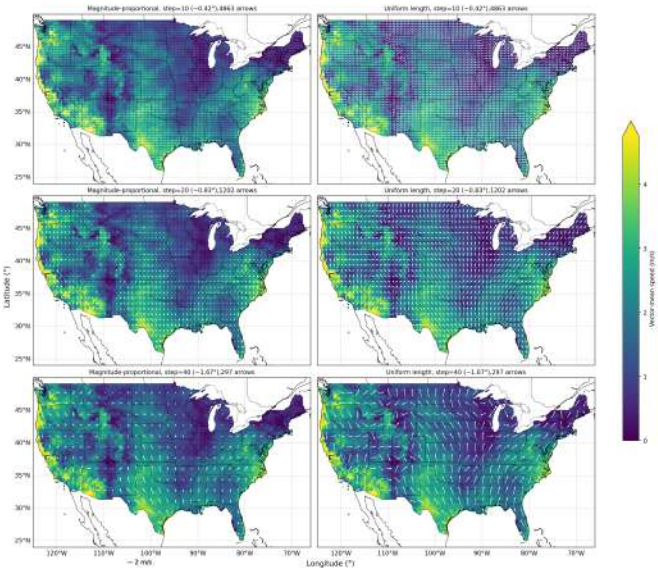


Fig. 17. Wind Velocity United States, 14-days mean starting 2021-07-01: Grid sampling and arrow comparison: Stride Method

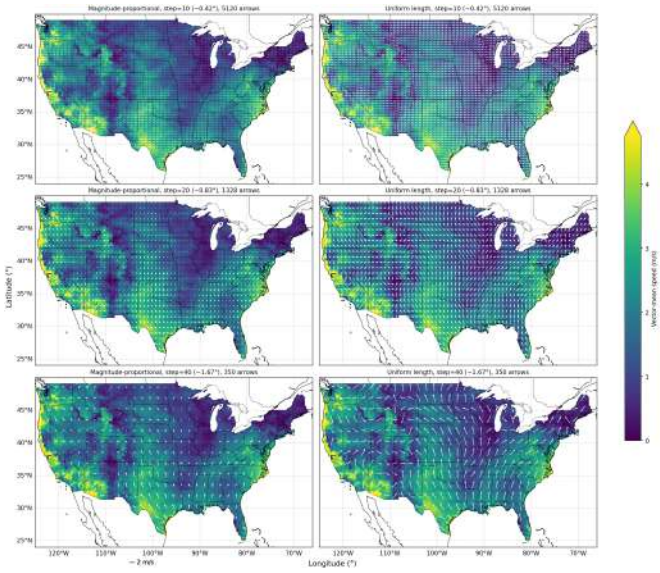


Fig. 18. Wind Velocity United States, 14-days mean starting 2021-07-01: Grid sampling and arrow comparison: Coarsen Method

6) Coarsen Grid Sampling : Coarsen sampling is a block sampling method where every arrow represents mean of a step x step block of cells.This is a smoother and more representative approach. The drawback of this approach is it looses finer details and when the windows within the block disagree, it results in weaker average wind vector.

Fig. 18 shows Biweekly mean wind velocity over united States starting from 1st July 2021 plotted in three different block sizes as quiver plots. For the three rows, block size is varied between every 10 x 10 , 20 X 20 and 40 X 40 block size from top row to bottom row respectively. The layout, color scale, and key are identical, but each arrow comes as mean wind velocity of entire block. The left column has arrows drawn proportional to the magnitude of average wind speed of the block along with colors representing wind speed redundantly while the right column has arrows drawn with uniform length and the magnitude of the wind is shown using colormap. The arrows are scaled with respect to space of the block or the step size. The arrows in the grid for 10, 20 and 40 steps are about 0.42°, 0.83°, and 1.67° spacing, with 5120, 1328, and 350 in total.

In case of step size 10, spacing of 10 X 10 block between 2 arrows, the map has highest no. of arrows. For the magnitude proportional arrow length, the arrows length is scaled according to block size, thus the arrow with highest wind speed occupying full space and weaker winds producing tiny arrows. The uniform length arrows are drawn as unit vectors with length 80% of the spacing of scale. As seen from figure, the smaller step size of 10 with magnitude proportional arrows produces dot like structure resulting in a sparse visualization while with uniform length arrows the visualization becomes very dense. In both cases it is very difficult to interpret any pattern. In case of block size 20, both the magnitude proportional and uniform arrow length maps are less sparse and less dense respectively compared to step size 10 and wind pattern is clearly visible. In case of step size 40, the magnitude proportional arrows map is sparser than step size 10 magnitude proportional arrows map but still tiny arrows for weaker winds make it difficult to identify an local features. For step size 40 uniform arrows map, the map is sparser than step size 20 uniform arrows map which make if difficult get any local details.

7) Comparison of Stride and Coarsen Grid Sampling: Stride method has fewer arrows than coarsen method at every step. A single sampled cell over ocean or outside the US is NaN and gets no arrow, whereas a coarsened block with even a few land cells still produces a value. In case of coarsen method the direction field is noisier, especially at step 40. Neighboring arrows differ more (the Inter mountain West and the Southeast), because one cell can be unrepresentative of its neighborhood. Map with Step 40 stride method is the cleanest and easiest to read, but small features are lost. In case of uniform length arrows,the flow pattern is seen clearly: southerly flow through the Plains and Gulf states, and a turn toward the east and northeast over the Mid-Atlantic. Weak-wind regions still show confident-looking arrows, so noisy directions look as reliable as strong ones.

III. Information Visualization

US Accidents [2] is a countrywide motor vehicle accidents dataset that covers 49 states of USA. It contains 2.25 millions traffic accident cases collected over a span of March 2019 to February 2023. The dataset has source, severity, time, location, weather conditions, road features and light conditions at the time of accidents. This dataset is used to perform the three information visualization tasks.

A. Parallel coordinates plot with edge bundling and zoomable treemap visualizations.

Treemaps and parallel coordinates plots were implemented using the US Accidents dataset to explore accident distributions and relationships between different attributes. Treemaps were used to represent hierarchical accident data, where rectangle size indicates accident count and colour distinguishes categories. Different hierarchies and spatial partitioning strategies were explored. Parallel coordinates plots were used to visualize multiple accident-related attributes simultaneously, with separate plots for weather conditions and road infrastructure. Severity was used for colour encoding, while brushing and filtering enabled interactive exploration of the data.

1) Dataset and Preprocessing: The tasks use the US Accidents dataset [2], which contains records of traffic accidents across the United States, along with information about their severity, location, time of occurrence, weather conditions and road infrastructure. The parallel coordinates plot were plotted using the variable: Severity,Temperature, Visbility,Precipitation, Wind speed,Bump, Crossing,Junction,Humidity and Traffic Signal.

Treemaps were plotted using the accident counts per location and based on the severity. The time of occurence was also considered by making use of the "Civil Twilight" attribute.

Preprocessing:

- Data Loading: The preprocessed US Accidents CSV file was loaded using the Pandas library.
- Column Selection: Only the relevant attributes required for the two visualization techniques were retained to reduce memory usage and processing time.
- Missing Value Handling: Records with missing values in the selected attributes were removed to ensure consistency during visualization.
- Data Sampling: A representative subset of accident records was selected to reduce visual clutter and improve browser rendering performance. Sampling was performed separately for each severity level to maintain representation across categories.
- Hierarchical Data Transformation for Treemaps: The tabular accident records were grouped and aggregated into hierarchical structures. Three different hierarchies were generated: State $\rightarrow$ City, State $\rightarrow$ Severity, and Sunrise$_{S}unsetßSeverity$.
- JSON Generation: The resulting hierarchical structures were converted into JSON files containing parent nodes, child nodes and accident counts. These files were used as input for the interactive treemap implementation.

2) Parallel Coordinates: Fig 19 shows the parallel coordinates plot over several severities. The visualization displays 10,000 accident records across Severity, Bump, Crossing, Junction and Traffic Signal.The infrastructure attributes exhibit concentrated values at 0 and 1 because they represent binary features.Despite bundling, considerable overlap remains because multiple records share identical or similar combinations of binary attributes.

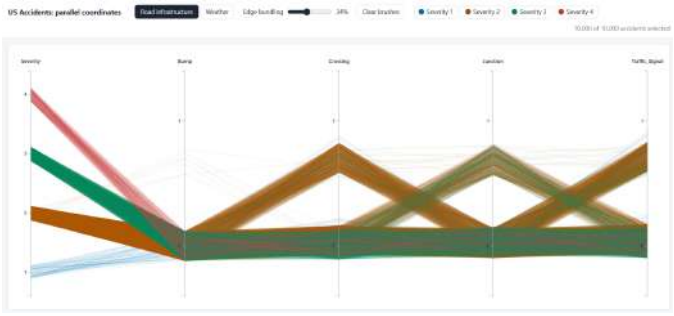


Fig. 19. Parallel coordinates plot for road infrastructure. Colour distinguishes severity



Fig. 20. Severity 4 accidents vs road infrastructure

Fig 20 represents severity 4 accidents vs the road infrastructure. The visualization displays the distribution of Severity 4 accidents across four road infrastructure attributes: Bump, Crossing, Junction and Traffic Signal.A large concentration of records is observed at zero across all four infrastructure axes, indicating that these features were not recorded as present for many accidents. This could also be due to the skewness of the data in the original dataset itself so we cannot conclude anything meaningful or reliable from this observation. Edge bundling at 58 percent produces thick, concentrated bands by grouping similar trajectories. This makes dominant patterns more visible.

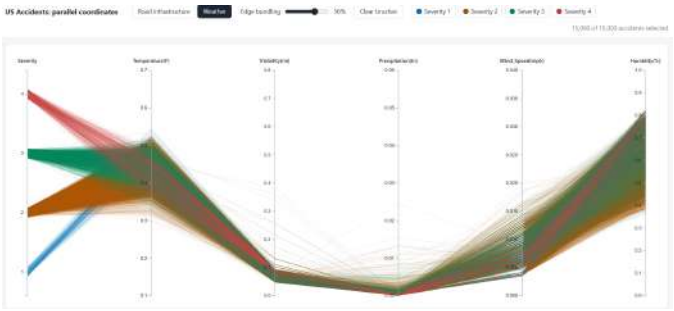


Fig. 21. Parallel coordinates plot for weather

Fig 21 represents weather attributes. The visualization represents five weather attributes: Temperature, Visibility, Precipitation, Wind Speed and Humidity, along with Sever-

ity.Temperature exhibits a broad distribution across accident records, with considerable overlap between severity categories.Visibility values are concentrated towards the lower end of the displayed range, while precipitation values are predominantly close to zero. However, precipitation values are also skewed in the dataset as most of the accidents that were recorded took place when there was no rain. Wind Speed and Humidity show considerable variation across the selected records.All 15,000 sampled records are displayed, providing an overview of the weather conditions associated with the different severity categories.

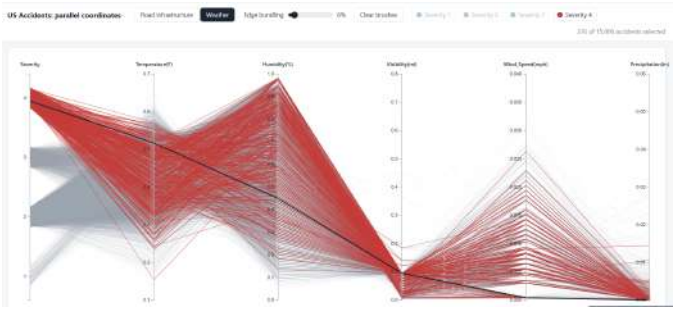


Fig. 22. Plot with bad ordering of attributes and no edge bundling

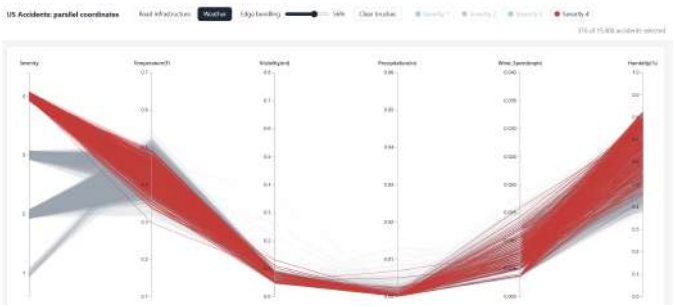


Fig. 23. Plot with good ordering and edge bundling

Figures 22 and 23 demonstrate the effect of edge bundling on the weather and road infrastructure visualizations, respectively. In the weather plot, a bundling strength of 56 percent combined with the revised attribute ordering (Temperature, Visibility, Precipitation, Wind Speed and Humidity) produces more concentrated groups of similar trajectories, making dominant patterns easier to identify. In the road infrastructure plot, a bundling strength of 34 percent helps group similar trajectories across the binary attributes (Bump, Crossing, Junction and Traffic Signal). However, considerable overlap remains because many accident records share identical or similar combinations of infrastructure features. Overall, edge bundling improves the visibility of common patterns, although excessive bundling can obscure individual variations.

3) TreeMaps: Fig 24 represents the geographical distribution of accidents across different US states, with each rectangle corresponding to a state and its area proportional to the



Fig. 24. Based on accident count

total number of accidents recorded in that state. California occupies the largest area, indicating the highest accident count, followed by states such as Florida, Texas, South Carolina and New York. Smaller rectangles represent states with comparatively fewer recorded accidents. The Squarified layout efficiently utilizes the available space while maintaining rectangles that are relatively easy to compare. This visualization provides an overview of the geographical concentration of accidents and allows users to identify states contributing a larger proportion of the total records.



Fig. 25. Slide and dice Tree Map

Fig 25 presents the State–City hierarchy using the Slice-and-Dice spatial partitioning strategy. Unlike the Squarified layout, which attempts to produce rectangles with more balanced aspect ratios, Slice-and-Dice divides the available space into alternating vertical strips. The resulting visualization contains several elongated and narrow rectangles, particularly for states with smaller accident counts. Although the area of each rectangle continues to represent the corresponding accident count, the varying aspect ratios make smaller regions more difficult to distinguish and compare. This experiment illustrates how the choice of spatial partitioning algorithm affects the readability and visual organization of a treemap.

Fig 26 shows a zoomed-in view of California from the State–City hierarchy, where individual rectangles represent



Fig. 26. Zoomed in treemap (California)

cities and their areas correspond to the number of recorded accidents. Los Angeles occupies the largest area, followed by cities such as Sacramento, San Diego, San Jose and Riverside. Several smaller rectangles represent cities with comparatively fewer accident records. The Squarified layout provides a more balanced arrangement of rectangles, allowing the larger cities to be identified easily while still accommodating a large number of smaller categories. The drill-down interaction demonstrates how treemaps support exploration from a national overview to more detailed city-level accident distributions.

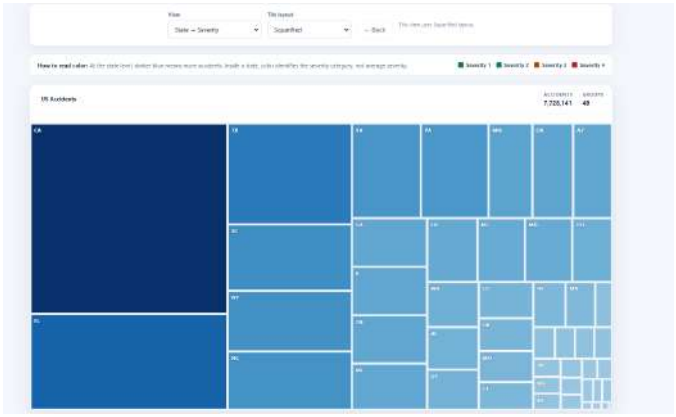


Fig. 27. Severity based treemap

Fig 27 represents the distribution of accident severity across different US states using the State–Severity hierarchy. At the state level, rectangle area and varying shades of blue represent the total number of accidents, with darker shades indicating higher accident counts. On selecting a state, its rectangle is divided into smaller rectangles representing Severity 1, 2, 3 and 4, with each category assigned a distinct colour. California occupies the largest area, followed by states such as Texas and Florida, indicating their comparatively higher accident counts. This hierarchical representation allows users to compare accident counts across states and subsequently examine the severity composition within individual states.

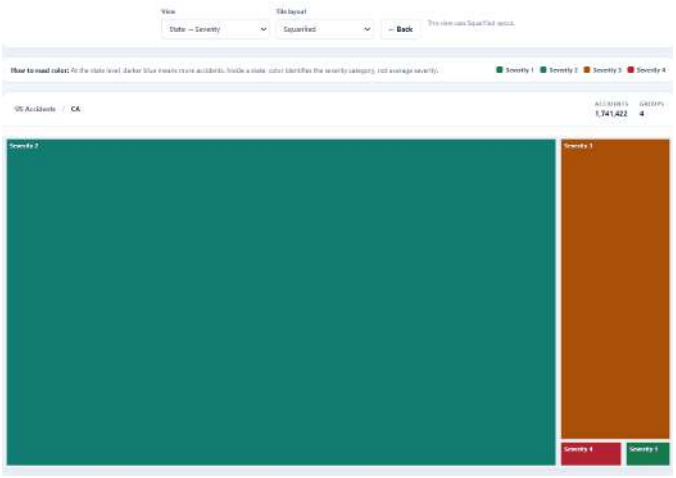


Fig. 28. Plot with good ordering and edge bundling

Fig 28 displays the distribution of accident severity within California after selecting the state from the State–Severity hierarchy. Severity 2 occupies the largest area, indicating that it accounts for the majority of recorded accidents in California, followed by Severity 3. Severity 1 and Severity 4 occupy considerably smaller areas. The Squarified layout allocates space according to the accident counts, making differences between severity categories easy to identify. This drill-down view demonstrates how the treemap can reveal the composition of accident severity within an individual geographical region.

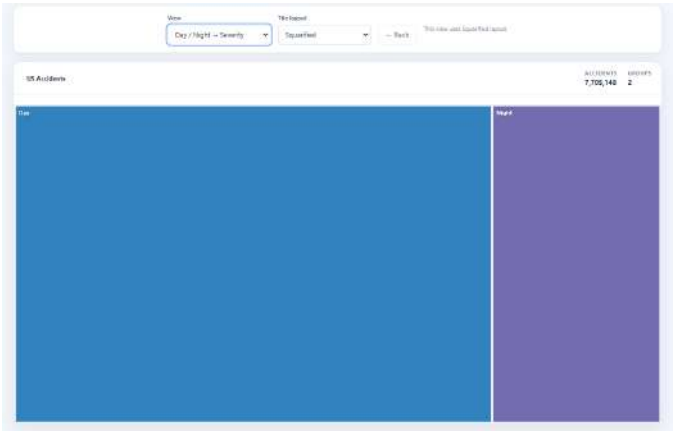


Fig. 29. Plot with good ordering and edge bundling

Fig 29 represents accident distribution across two categories: Day and Night, based on the Civil Twilight attribute. The area of each rectangle corresponds to the number of accidents recorded during the respective period. The Day category occupies a considerably larger area than Night, indicating that a greater number of accidents in the dataset occurred during daylight conditions. The Squarified layout makes the difference

in accident counts between the two categories immediately visible. Selecting either rectangle allows users to explore the corresponding accident severity distribution in greater detail.

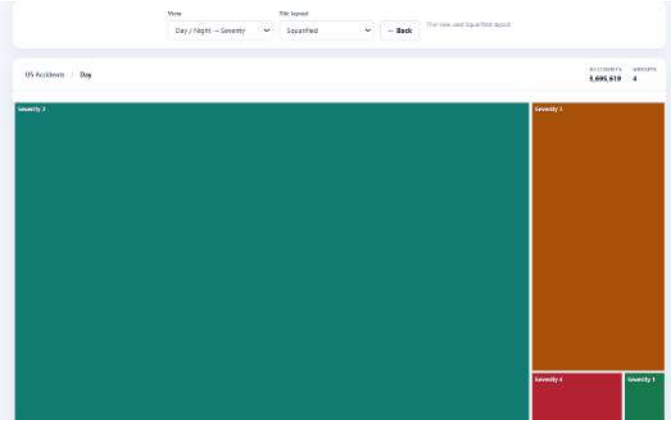


Fig. 30. Day-Time severity distribution

This figure (Fig 30) shows the severity distribution of accidents recorded during daytime, obtained by drilling down into the Day category of the previous treemap. Severity 2 occupies the largest area, followed by Severity 3, while Severity 4 and Severity 1 account for considerably smaller proportions. The rectangle areas represent the number of accidents belonging to each severity category. The visualization reveals that Severity 2 accidents dominate the daytime records in terms of absolute count. The interactive drill-down functionality makes it possible to examine the severity composition of individual time-of-day categories.

B. Summary

The parallel coordinates visualizations reveal variations in weather conditions and road infrastructure features across different accident severity categories. Weather attributes such as temperature, visibility, precipitation, wind speed and humidity exhibit considerable overlap across severity levels, suggesting that no individual attribute clearly distinguishes accident severity in the selected sample. The infrastructure plots reveal concentrated bands due to the binary nature of attributes such as Bump, Crossing, Junction and Traffic Signal, with many records sharing similar combinations. Experiments with attribute ordering and edge bundling demonstrate that rearranging axes and grouping similar trajectories can improve readability and reveal dominant patterns. However, excessive bundling can obscure individual variations. Interactive brushing and severity filtering further facilitate the exploration of specific subsets of accident records.

The treemap visualizations highlight geographical and temporal accident distributions, with California contributing the highest accident count and daytime records substantially outnumbering nighttime records. Severity 2 forms the largest category in the explored breakdowns. Comparing Squarified and Slice-and-Dice layouts demonstrates how spatial partitioning affects space utilization and readability, while interactive drill-down enables detailed state- and city-level exploration.

C. horizon plots and ridgeline plots

Horizon plots and ridgeline plots are two ways of comparing many time-dependent series or distributions at once. A horizon plot lets tens of series be stacked and compared in the space of one ordinary line chart. A ridgeline plot can be used to compare many distributionsalong a common axis. Horizon plots are used here for the *time-series* view of the US Accidents data (counts per month or per year), and ridgeline plots for the *group-wise distribution* view (when in the day, the year, or under what weather accidents occur, and how this differs between groups).

1) Dataset: The tasks use the US Accidents dataset [2] Each record is one traffic accident with its start time, location (state, city), a severity label, and the weather measured at the nearest airport station. Only the 16 columns needed for the two plot types were loaded: `ID`, `Source`, `Severity`, `Start_Time`, `Distance`, `City`, `State`, `Airport_Code`, `Weather_Timestamp`, `Temperature`, `Humidity`, `Visibility`, `Wind_Speed`, `Precipitation`, `Weather_Condition` and `Civil_Twilight`. The variables used by the final figures are:

- `Start_Time`: converted to month, year, day of week, day of year and a continuous hour of day (hour + minute/60).
- `State`
- `Weather_Condition`
- `Severity`
- `Visibility(mi)` and `Wind_Speed(mph)`

2) Data Preprocessing: The raw file has no duplicate IDs and no unparseable timestamps. Missing values were highest for precipitation (28.5%) and wind speed (7.4%), and about 2% for temperature, humidity, visibility and weather condition.

Cleaning. Rows with a missing start time, state or severity were dropped, severity was checked to lie in $\{1, 2, 3, 4\}$, and state and city names were trimmed and upper-cased.

Weather condition strings were mapped once (on the unique strings, not on millions of rows) to keyword-based groups: Thunderstorm, Snow (snow, sleet, hail, ice pellets), Rain (rain, drizzle, showers), Fog (fog, mist, haze, smoke), Cloudy, Clear, Unknown (missing condition) and Other.

Numeric fields outside plausible limits were set to missing: temperature outside $[-40, 130]$ °F, humidity outside $[0, 100]\%$, visibility outside $[0, 20]$ mi, wind outside $[0, 120]$ mph, precipitation outside $[0, 10]$ in and distance outside $[0, 50]$ mi.

A missing or `UNKNOWN` station code was replaced by the most frequent station of the same city, then of the same state. A missing reading was then replaced by the nearest-in-time valid reading of the same station. Where the weather timestamp itself was missing the accident start time was used instead. This reduced the missing share significantly.

Analysis window. To avoid partial years that would look like artificial rises or falls, only the full calendar years 2017–2022 were analysed

3) Methodology: for horizon and ridge plots

Horizon plots. Time series were split into positive and negative deviations from the median baseline and displayed using multiple color bands. Positive values are shown in red and negative values in blue, with darker shades indicating larger deviations. A shared band height was used across all rows for consistent comparison, and rows were ordered using hierarchical clustering. To reduce the effect of the long-term increase in accident counts, three versions were compared: raw counts, counts normalized by the series median, and counts divided by a 12-month moving average (detrended ratio).

Ridgeline plots. Ridgeline plots were created using smoothed kernel density estimates for each group. Circular smoothing was applied for hour-of-day and day-of-year variables. Each density was normalized to represent the distribution rather than the number of accidents. Groups were arranged in their natural order, with sample size shown for each ridge, while median and mean values were indicated where applicable.

TABLE I
SPECIFICATION OF THE FINAL PLOTS.

Plot	Rows (groups)	Axis / granularity	Setting
Horizon A	top-10 states	monthly, 72	detrended, similarity order
Horizon B	7 weather groups	monthly, 72	detrended, similarity order
Horizon C	top-20 states	yearly, 6	index, order by peak year
Horizon D	6 years	month of year	detrended ratio, chronological
Ridgeline A	7 weekdays	hour of day	bw 0.4 h, circular
Ridgeline B	6 years	day of year	bw 8 days, shared scale
Ridgeline C	4 severities	visibility, wind	bw 0.4 mi / 1 mph

4) Experiments: One setting was varied at a time, with all others fixed, to choose the final design. Table I lists the final plots. Variations in group and granularity are covered by monthly vs. yearly counts, states vs. weather groups, and a calendar vs. month-of-year layout (horizon), and by weekday, year and severity groups (ridgeline).

Number of bands. Two, four and six bands were compared on the four largest states. Two bands show only extreme months. Six bands give shades that are hard to tell apart. Four bands were chosen.

Normalisation. Raw counts, a relative index (value / median) and a detrended ratio (value / 12-month average) were compared for the top 10 states. Raw counts are dominated by California and Florida. The relative index is dominated by growth and the late-2020 level shifts in Florida, Virginia and Pennsylvania. The detrended ratio removes both and makes states comparable, so it was used for the monthly plots. The relative index was kept for the yearly plot, where growth is the topic.

Row order. Rows were ordered by total count, month of peak and similarity. Total count only repeats the totals. Similarity ordering places similar states together, so shared episodes form horizontal blocks. It gives the order MN, OR, TX, NY, SC, NC, CA, FL, VA, PA and was used.

Kernel bandwidth. Bandwidths of 0.2, 0.4 and 0.8 h were compared for the hour-of-day density. At 0.2 h the curves are jagged. At 0.8 h the peaks flatten. A bandwidth of 0.4 h keeps both adequately.

Ridge scaling. Severity ridges were drawn on their own peak and on one common density scale. Each density integrates to one, so the common scale shows concentration, not counts (counts are the n in the labels). Severity 1 is sharply double-peaked, while Severity 4 is flat and spread over the day. Own-peak scaling was used to compare shapes (Ridgeline A and C), and the common scale to compare years (Ridgeline B).

5) Horizon Plots: The final horizon plots use four bands, the median as baseline and a shared band height for all rows. Detrended ratios were used for the monthly plots and the relative index for the yearly plot. Band heights are in the units of the ratio.

Horizon A: monthly accidents, top 10 states. Each row is one of the ten states with the most accidents, shown month by month for 2017–2022 (detrended), with similar states placed next to each other.

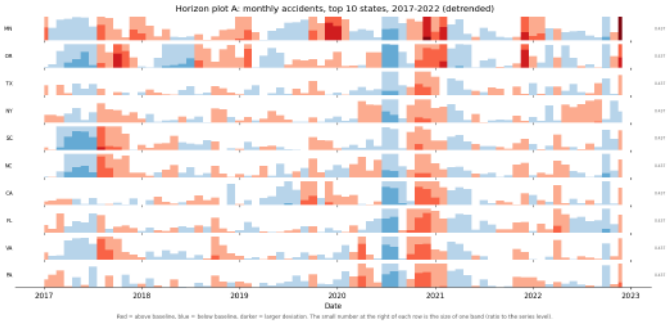


Fig. 31. Horizon plot A: monthly accidents of the top 10 states, 2017–2022, detrended. Red = above the state's median, blue = below, darker = larger deviation; one band = 0.437 of the 12-month average.

Observations (Horizon A)

- Most states show a clear drop in accidents around mid-2020, followed by an increase from late 2020 to early 2021.
- Minnesota and Oregon show the strongest seasonal pattern, with accident counts increasing during the winter months. California, Texas, and Florida show much smaller seasonal changes.
- Florida shows a noticeable increase in accident counts after mid-2020, indicating a long-term change rather than a seasonal effect.
- New York differs from the other states, with its highest accident counts occurring around September instead of winter.
- Some states show unusually low accident counts during early 2017, which is likely due to incomplete data coverage rather than actual traffic conditions.
- The detrended plots show positive correlation between all states, suggesting that they share similar seasonal patterns while still having state-specific variations.

Horizon B: monthly accidents by weather group. Same monthly, detrended layout as Horizon A, but each row is a weather group instead of a state.

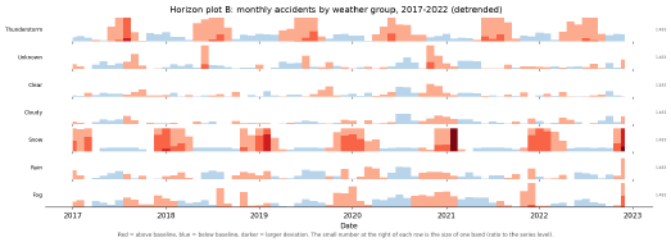


Fig. 32. Horizon plot B: monthly accidents by weather group, 2017–2022, detrended. Unknown = missing `Weather_Condition`. One band = 1.483 of the 12-month average.

Observations (Horizon B)

- Snow shows the strongest seasonal pattern, with accident counts peaking every winter and remaining very low during summer.
- Thunderstorm-related accidents are highest during summer (May–September), showing the opposite trend to snow.
- Fog and rain show moderate seasonal variations, with higher accident counts during late autumn and winter.
- Clear and cloudy weather account for most accidents and show relatively small seasonal variation because these conditions occur throughout the year.
- The `Unknown` weather category shows an unusual increase around late 2020, likely due to changes in data reporting rather than actual weather conditions.
- These plots show when different weather conditions are associated with accidents, but they do not indicate which weather condition is more dangerous.

Horizon C: yearly accidents, top 20 states. A coarser time unit: one bar per year for the top 20 states, relative to each state's median year and ordered by the year of the peak.

Observations (Horizon C)

- Most states show lower yearly accident counts in 2017–2018 and higher counts in 2021–2022, indicating an overall increase over the study period.
- States such as Virginia, Pennsylvania, Florida, and New Jersey show the largest growth, while Texas, Illinois, and Georgia remain relatively stable across the years.

Horizon D: accidents by month of the year. The detrended national monthly series is folded by year: each row is one year and the horizontal axis is the month of the year, so it shows which months are above or below normal in each year.

Observations (Horizon D)

- The month-of-year horizon plot shows a pronounced drop during July–September 2020 followed by a sharp increase towards the end of 2020, which is consistent with the change in data coverage.

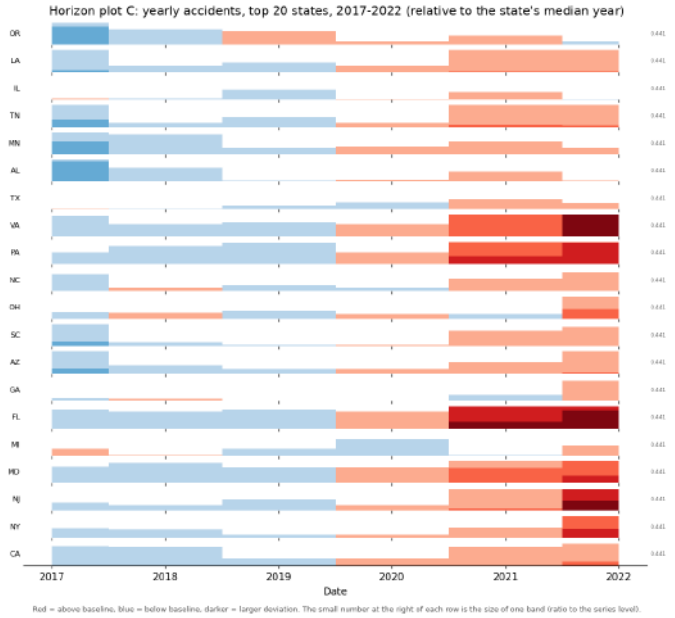


Fig. 33. Horizon plot C: yearly accidents of the top 20 states relative to the state's median year (one band = 0.441), ordered by peak year; each bar is centred on the January tick of its year.

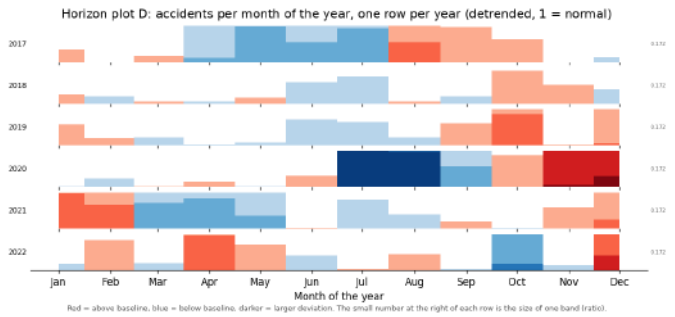


Fig. 34. Horizon plot D: accidents per month of the year, one row per year, detrended (1 = normal; one band = 0.172).

- Most years show higher accident counts during late autumn and winter, whereas summer months generally remain below the yearly average.

6) Ridgeline Plots: The final ridgelines use a bandwidth of 0.4 h for hour of day (circular), 8 days for day of year (circular), 0.4 mi for visibility and 1 mph for wind speed.

Ridgeline A: hour of day by day of the week. One ridge per weekday, each scaled to its own peak.

Observations (Ridgeline A)

- Monday to Thursday show similar patterns, with two clear peaks during the morning and evening commuting hours and lower accident density around midday.
- Friday also shows two peaks, but the evening peak is more spread out, indicating extended travel activity later in the day.
- Saturday and Sunday exhibit a single broad afternoon peak with a larger late-night tail, reflecting leisure and

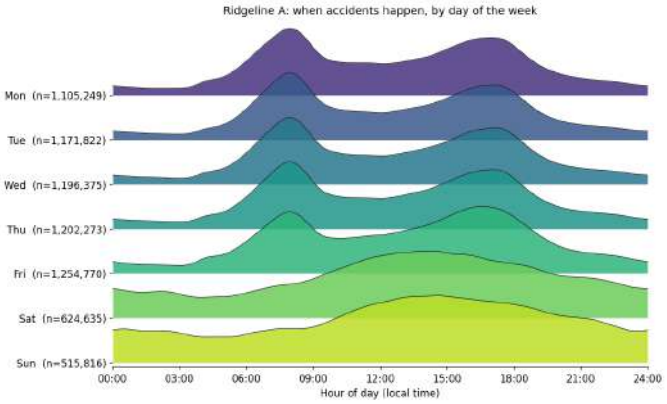


Fig. 35. Ridgeline A: density of the hour of day of accidents, one ridge per weekday (each scaled to its own peak).

weekend travel rather than commuting.

- Weekdays account for most accidents, averaging about 3,789 accidents per day compared to 1,822 on weekends. Since ridgeline plots show normalized densities, the sample size (n) is included for each day.

Ridgeline B: day of year by year. One ridge per year on a common density scale, so that the heights are comparable. The densities integrate to one in each year, so the growth of the total count does not distort the comparison. A flat ridge means that accidents are spread evenly over the year.

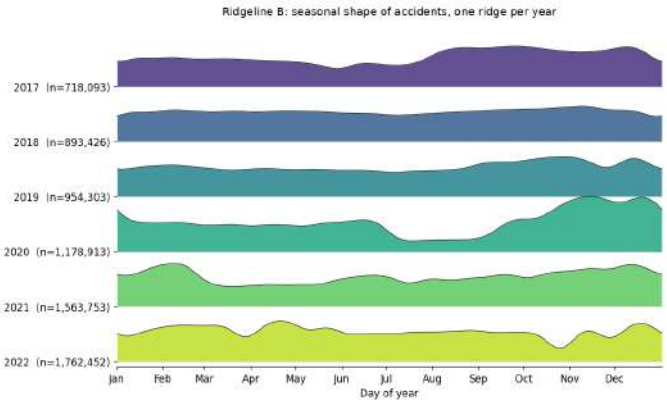


Fig. 36. Ridgeline B: density of the day of year, one ridge per year (common scale).

Observations (Ridgeline B)

- The ridgeline plots show relatively stable accident distributions in 2018 and 2019, with only a slight increase towards October–November.
- The 2017 distribution gradually increases from mid-year onwards, likely reflecting incomplete data coverage rather than a seasonal trend.
- The 2020 distribution shows a sharp decline during July–September followed by a strong increase towards the end of the year, consistent with changes in data coverage.
- The distributions vary across years, indicating that accident patterns are not identical every year. However, most years show a gradual increase towards late autumn and winter.

Ridgeline C: visibility and wind speed by severity. Distributions of two weather variables for the four severity levels, with the median (bar) and mean (diamond) marked.

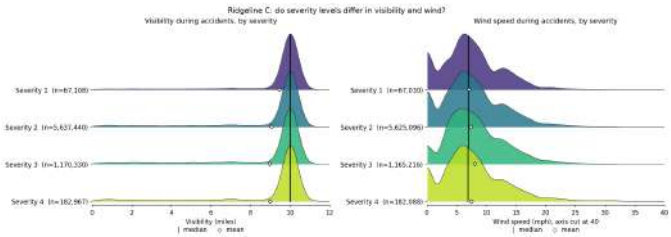


Fig. 37. Ridgeline C: distribution of visibility (left) and wind speed (right, axis cut at 40 mph) during accidents, one ridge per severity level. The vertical bar is the median and the diamond the mean.

Observations (Ridgeline C)

- Visibility distributions are similar across all severity levels, with most accidents occurring under good visibility (around 10 miles). Higher severity accidents show a slightly larger proportion of low-visibility conditions.
- Wind speed distributions are also similar for all severity levels, with most accidents occurring at low to moderate wind speeds (around 6–7 mph).
- A small spike near the maximum wind speed is caused by data capping and does not represent an actual weather pattern.
- Overall, accident severity shows only a weak relationship with visibility and wind speed. These plots describe weather conditions during accidents and do not indicate whether poor weather increases accident risk.

7) Discussion: Horizon plots let 10 to 20 series be compared in a few centimetres of height. The choice of normalisation mattered most: raw counts show only California and Florida, the relative index shows growth and level shifts, and the detrended ratio shows season and shared events. Two kinds of structure appear. The first is real seasonality: winter peaks in Minnesota and Oregon, snow in winter and thunderstorms in summer, and a weak late-autumn/winter rise nationally (Horizon D). The second is data collection: the mid-2020 drop across all states, the late-2020 level shifts in Florida, Virginia and Pennsylvania, and the `Unknown` weather peak. Because these appear in many series at the same time, they are more likely artefacts of the data than properties of individual states.

Ridgeline plots show distribution shapes that single numbers hide. The weekday curves have two commuting peaks (about 07:55 and 16:55) and the weekend curves a single afternoon hump around 14:15 (Ridgeline A), so the day of the week changes the *shape* of the daily rhythm and not just its level. Visibility and wind speed (Ridgeline C) differ very little between severity levels, so severity is only weakly tied to

these two variables. The experiments show that bandwidth and scaling must be chosen with the question in mind: a small bandwidth adds noise, a large one hides detail, and the common scale shows concentration while own-peak scaling shows shape.

D. Node-link diagrams

Node-link diagram is a graph visualization where a network is drawn using dots and lines. The dots are nodes which can be cities,people, states and the lines are the relationships between them. The US Accidents dataset is first transformed into a network dataset and then three different graph layout algorithms Spring layout, Kamada kawai layout and Tutte barycentric layout are used to draw node-link diagrams for the US Accidents dataset.

1) Preprocessing: To transform the US Accidents dataset into network dataset, it is necessary to identify the nodes and relationship between them in US Accidents dataset. As the US Accidents dataset contains motor accidents data across all 49 states of USA and it has record of location of accident, this spatial information is used to form the nodes and edges. Cities and states are identified as nodes. The accident records are grouped by state wise and within state city wise. There are two types of edges formed. There is an edge between state and the city within it. Another types of edge is edge between two states. To form the edges between states, first the centroid of every state is calculated and the k closest neighbors based on centroid for every state are identified. The mutual state <-> state neighboring pairs are retained as network edges. These edges and nodes are used to draw graph visualizations.

2) Implementation: NetworkX python library provides capabilities to create,analyze and draw graphs. It is used to build the structure of US Accidents network and to draw and analyze the graph. Matplotlib python library was also used along with networkX for plotting the graph. Ipympl is python package that lets matplotlib figures be interactive. This Iypympl is used to convert static US Accidents graph into interactive widget and user interactions are added to each graph. User interactions help the user to select a particular area of the visualization to inspect in greater detail.

3) Spring Layout: Spring layout algorithm is force directed algorithm which is inspired from physics spring law where the links act as springs pulling connected nodes together and all nodes of same type repel each other. It starts with assigning random positions to nodes and converges after certain iterations when the forces balance. It is another name to Fruchterman-Reingold algorithm. Fig. 38 shows US Accidents network using spring layout.

Here states are positioned as hubs, cities are spiraled around hubs and the accident count is used as feature encoding for both. First a graph of states as hubs at random positions and their border links is drawn using spring layout force-directed algorithm. 300 iterations are used for position to settle. For every state, cities are sorted with accident count and only top 10 cities are chosen to draw around it's state in circle. Every state and it's cities form a cluster and the algorithm takes care

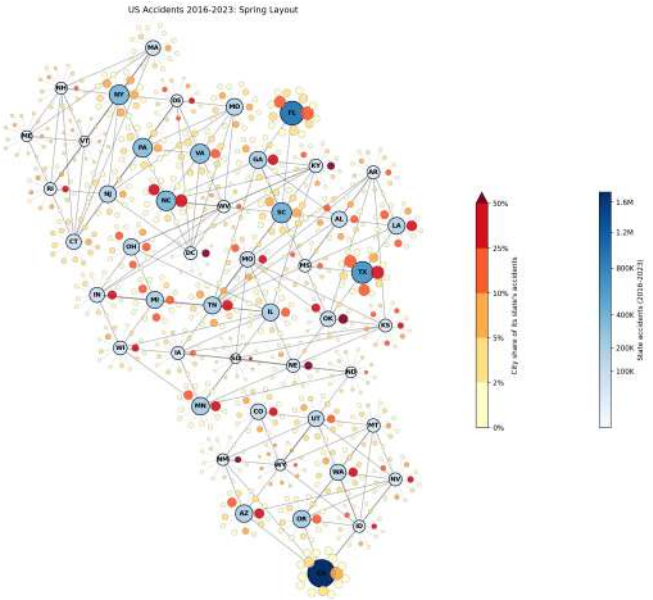


Fig. 38. US Accidents Network Spring Layout

that clusters do not overlap. Two different color bars are used one for states and one for cities. For states it is continuous colorbar where color indicating total no of accidents in that state. Size of the circle also shows no. of accidents in the state redundantly. The size of the cities show the raw accident count in that city while the color of the city shows the share of city in the total accidents in it's state. Only state hubs are labeled. This graph is provided with a user interaction where user can zoom on particular state by clicking the state hub. Fig. 39 shows an example.

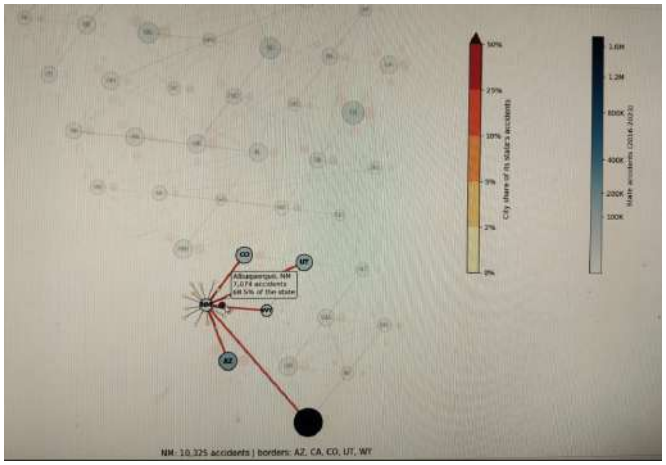


Fig. 39. US Accidents Network Spring Layout User Interaction

In Fig. 39 user has selected state New Mexico by clicking on NM hub. The state NM, its cities and top k connected neighbors are highlighted and the rest of the graph is grayed showing the focus on area of interest. It also shows the number of accidents happened in state New Mexico and top k nearest neighboring states. The red color of the city indicates it is the one with highest share in no of accidents happened in the state

thus being hotspot city for accidents. The tooltip interaction shows the city name Albuquerque, total no of accidents in the city and its share 68.5% in state total accidents.

4) Kamada Kawai Layout: Kamada kawai layout is another force directed graph layout algorithm where graph theoretical distance between nodes is maintained in the plot also. Shortest path distance between every pair of nodes is calculated and link between them is treated as spring whose length is proportional to shortest distance. Nodes are moved untill the total spring energy is as low as possible. Fig. 40 shows the US Accidents network in Kamada kawai layout.

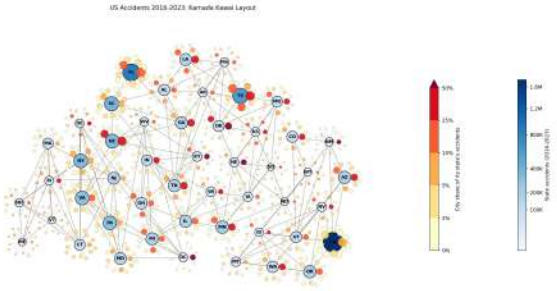


Fig. 40. US Accidents Network Kamada Kawai Layout

Here each state is a hub and top 15 cities within the state with highest accident count are placed as dots around each hub. First a small graph of states with only border links is created. Kamda kawai treats every pair of states connected by string whose length is proportional to the number of actual border hops between them thus the states who share borders are nearer and other are far apart if there are many borders between them. The algorithm s deterministic so every run gives same hub arrangement. Every state and it's cities form a cluster. Two different color bars are used one for states and one for cities. For states it is continuous colorbar where color indicating total no of accidents in that state. Size of the circle also shows no. of accidents in the state redundantly. The size of the cities show the raw accident count in that city while the color of the city shows the share of city in the total accidents in it's state. Only state hubs are labeled. This graph is provided with a user interaction where user can zoom on particular state by clicking the state hub. Fig. 41 shows an example. In Fig. 41 user has selected state Illinois by clicking on IL hub. The state IL, its cities and 7 closest connected neighbors are highlighted and the rest of the graph is grayed showing the focus on area of interest. It also shows the number of accidents happened in state Illinois and top 7 nearest neighboring states. The faint yellow color of the city, whose tooltip is shown, indicates it is the one with very low share in no of accidents happened in the state. The tooltip interaction shows the city name Willowbrook, total no of accidents in the city and its share 1.2% in state total accidents.

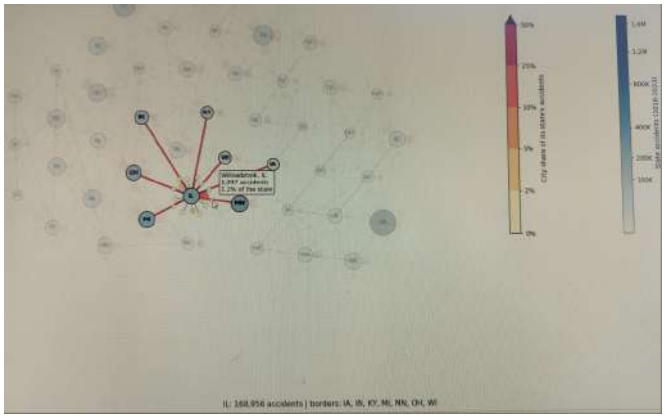


Fig. 41. US Accidents Network Kamada Kawai Layout User Interaction

5) Tutte Barycentric Layout: Tutte Barycentric layout is a method of drawing planar graph with no crossing links. Tutte barycentric algorithm picks one face of the planar graph and puts all its vertices at the corners of convex polygon. Every other node of the graph is then placed at the barycentre of its neighbors' positions in the graph. There are no iterations and no random start at there is only one linear equation to solve the positions of internal nodes with respect to positions of nodes at the corners. Fig. 42 shows the US Accidents network in Tutte Barycentric layout.

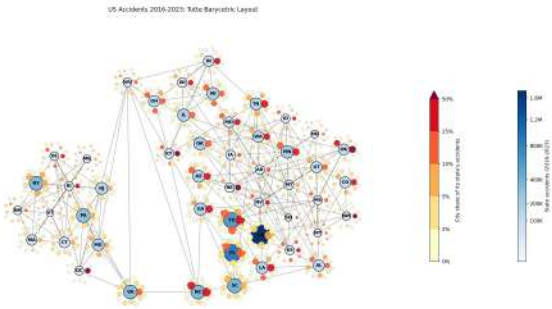


Fig. 42. US Accidents Tutte Barycentric Layout

As Tutte barycentric layout works only with planar graphs the US Accidents network is converted to planar graph. The planar graph algorithm takes out one edge every iteration and checks the remaining graph for planarity. The longest edge is always removed first. To determine the longest edge haversine distance between state nodes is used. The resulting graph has no-crossings. Tutte barycentric embeddings stores the cyclic order of neighbors for each node , which describes how the whole graph sits in one plane. In the graph, all the outer state nodes which are hubs have formed a convex polygon. Every state and it's cities circled around it form a cluster. The top 15 cities with very high accident count are selected for drawing. Algorithm takes care of overlapping clusters at

every iteration. Two different color bars are used one for states and one for cities. For states it is continuous colorbar where color indicating total no of accidents in that state. Size of the circle also shows no. of accidents in the state redundantly. The size of the cities show the raw accident count in that city while the color of the city shows the share of city in the total accidents in it's state. Only state hubs are labeled. This graph is provided with a user interaction where user can zoom on particular state by clicking the state hub. Fig. 43 shows an example.

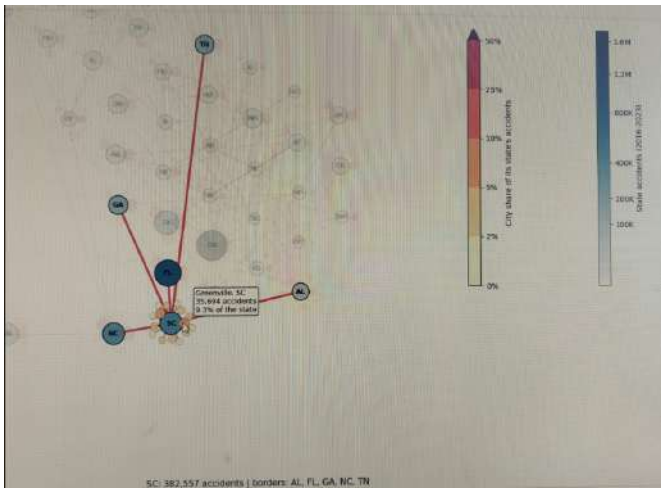


Fig. 43. US Accidents Network Tutte Barycentric Layout User Interaction

In Fig. 43 user has selected state south Carolina by clicking on SC hub. The state SC, its cities and 5 closest connected neighbors are highlighted and the rest of the graph is grayed showing the focus on area of interest. It also shows the number of accidents happened in state South Carolina and names of 5 nearest neighboring states. The orange color of the city, whose tooltip is shown, indicates it is the one with low share in no. of accidents happened in the state. The tooltip interaction shows the city name GreenVille, total no. of accidents in the city and its share 9.3% in state total accidents.

6) Comparison of Three graph layouts: The kamada kawai layout algorithm for graph visualization is more aesthetically pleasing than spring layout and Tutte barycentric layout algorithms. The spring layout algorithm starts with random positioning of nodes and then over the iteration the places are stabilized. During the process of stabilization and removing cluster overlaps the overall graph can result stretched as seen in Fig. 38. Without seed, for every run the graph may result in different arrangement of nodes. As kamada kawai algorithm is based on the theoretical distances between two nodes of the graph, the arrangement does not need iterations and every run produces same arrangement of nodes. This layout is often even, symmetric and has less tangled links than spring layout as can be seen from Fig. 40. The Tutte barycentric layout algorithm needs planar graph so the original graph needs to be altered to remove cross links. This may result into disconnect components. The layout depends on which planar face is selected first to form convex polygon and how many vertices it has. The positions of other vertices is based on positions of neighbors starting with nodes having neighbors on the corners of the polygon. This can result in uneven graph as can be seen in Fig. 42. For US Accidents dataset the kamada kawai layout algorithm looks better suited as it has produced aesthetically pleasant visualization.

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